REFUTATION OF BOTH BRANCHES OF ERDŐS PROBLEM #43 ON DISJOINT-DIFFERENCE SIDON SETS

ARGUS OPEN RESEARCH PROJECT

Abstract. Let $f(N)$ denote the maximum cardinality of a Sidon subset of $\{1,\dots,N\}$. Erdős Problem #43 asks whether every pair of Sidon sets $A, B \subseteq \{1,\dots,N\}$ with $(A-A)\cap(B-B)=\{0\}$ satisfies $\binom{|A|}{2}+\binom{|B|}{2} \le \binom{f(N)}{2}+O(1)$, and whether adding $|A|=|B|$ forces $\binom{|A|}{2}+\binom{|B|}{2} \le (1-c+o(1))\binom{f(N)}{2}$ for some absolute constant $c>0$. We document the refutation of both branches, including the complete formal verification of the compact Cayley clique theorem for Problem #42/#43 in Lean 4.27.

1. Main Results

Theorem 1.1. *Both branches of Erdős Problem #43 are false:*

(1) Unbounded additive excess: *the negative resolution of Erdős Problem #42 (formally verified in Lean 4.27 using only classical foundation axioms) produces for every target size k and all sufficiently large N a maximal Sidon set $A \subseteq \{1,\dots,N\}$ of cardinality $f(N)$ together with a Sidon set $B \subseteq \{1,\dots,N\}$ of cardinality $|B|=k$ satisfying $(A-A)\cap(B-B)=\{0\}$, so that $\binom{|A|}{2}+\binom{|B|}{2}-\binom{f(N)}{2}=\binom{k}{2}$ is unbounded.*

(2) Equal-size branch ($|A|=|B|$): *Barreto's difference-set projection constructs equal-size Sidon sets $A, B \subseteq \{1,\dots,N\}$ with $(A-A)\cap(B-B)=\{0\}$ and $|A|=|B|=(1/\sqrt{2}-o(1))\sqrt{N}=(1/\sqrt{2}-o(1))f(N)$, yielding $\binom{|A|}{2}+\binom{|B|}{2}=(1-o(1))\binom{f(N)}{2}$ and ruling out any positive constant saving $c>0$.*

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